Generative Vehicle Trajectory Prediction with Diffusion and Flow Matching

Machine Learning Course Project Proposal

Abstract. Vehicle motion is intrinsically multi-modal: after the same observed history, a driver may continue, turn, yield, or change lanes. This project proposes a controlled comparison of deterministic regression, diffusion modeling, and flow matching for conditional trajectory prediction. By sharing data, conditioning, and network capacity across methods, we will test whether generative models better cover plausible futures and whether flow matching can retain predictive quality with fewer sampling steps.

1. Project Idea and Research Questions

Single-output regression trained with mean squared error often averages incompatible futures and may produce an implausible "mean trajectory." Generative models instead approximate a conditional distribution and permit multiple samples. We ask: (RQ1) Do diffusion and flow matching improve best-of-K trajectory accuracy over deterministic regression, especially at intersections? (RQ2) Can flow matching approach diffusion accuracy with lower sampling latency? (RQ3) How do sample count and integration steps affect accuracy, diversity, repeatability, and runtime? Conclusions will be empirical; no superiority claim is assumed in advance.

2. Implementation Plan

Data and task. We will use the nuPlan mini split and formulate an agent-centric prediction task. Approximately two seconds of target-agent history, nearby-agent context, and available lane/map information will condition an eight-second future trajectory sampled at 10 Hz. Positions will be transformed into the target vehicle's local coordinate frame. To control storage and runtime, scene features and trajectory pairs will be validated on a small subset and cached before scaling to the feasible training set.

Models. A deterministic MLP trajectory regressor will establish the baseline. The generative systems will use the same context encoder and a compact time-conditioned denoising backbone. The diffusion model will learn noise prediction under a fixed variance schedule and sample using a reduced-step solver. Flow matching will learn the conditional velocity field along a linear probability path and sample by Euler integration of the corresponding ODE. Parameter counts, data splits, conditioning features, training epochs, and approximate compute budgets will be matched to make the comparison defensible.

Evaluation and experiments. For K sampled futures, we will report minADE@K, minFDE@K, and miss rate, supplemented by inference latency, number of function evaluations, and a simple diversity measure. Core experiments vary $K \in \{1, 4, 6, 16\}$ and compare diffusion sampling steps with 4/8-step flow integration. A fixed-noise repeatability test will verify deterministic ODE sampling behavior. Bird's-eye-view visualizations will examine intersection, turning, straight-road, and lane-change cases; failure analysis will identify off-road, collapsed, or physically implausible samples. Where computationally feasible, results will be averaged over multiple seeds.

Expected outputs and risks. Deliverables include a cached nuPlan preprocessing pipeline, three reproducible models, metric/runtime tables, step and sample-count ablations, trajectory visualizations, failure cases, and a final report. Major risks are complex nuPlan preprocessing, limited compute, unstable training, and an unfair model comparison. Early preprocessing validation, shared architecture components, reduced data/step settings, checkpoints, and a baseline-first schedule will preserve a complete core result even if extensions are constrained.

3. Group Members' Responsibilities

Member A	nuPlan extraction, coordinate transforms, scene features, cached splits, and data validation.
Member B	Shared context encoder, deterministic baseline, metrics, and common evaluation harness.
Member C	Diffusion objective, training, reduced-step solver, stability checks, and compute logging.
Member D	Flow-matching objective and ODE sampler, cross-model ablations, visualization, and report integration.

Key references: Caesar et al., "nuPlan," CVPRW 2021; Yang et al., "Diffusion-ES," CVPR 2024; Zheng et al., "Diffusion-Based Planning for Autonomous Driving," ICLR 2025.